University of Toronto
Faculty Of Applied Science & Engineering
Applied Fundamentals of Deep Learning

Date : April 29th, 2023
Duration: **150 minutes**
Aids Allowed: **Open Book & Non-Programmable Calculators**

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Do **NOT** turn this page until you have received the signal to start.

Do **NOT** write anything over the **QR Codes** on top of the pages.

This test consists of 9 questions on 18 pages (including this one), printed on both sides of the paper. When you receive the signal to start, please make sure that your copy of the test is complete, fill in the identification section above. Answer each question directly on the test paper, in the space provided. Make sure to answer the questions in their designated place.
Write up your solutions carefully! Marks cannot be awarded for solutions that are not understandable by the grader, and may be deducted if you make false assertions. If you are giving only one part of an answer, indicate clearly what you are doing. Part marks might be given for incomplete solutions.

1: _____/10
2: _____/10
3: _____/10
4: _____/10
5: _____/10
6: _____/10
7: _____/10
8: _____/10
9: _____/10

TOTAL: _____/90

Good Luck!

OVER...

Question 1. [10 MARKS]

Circle the best answer for each of the questions below. Do not circle more than one answer per question.

Part (a) [1 MARK]

How many parameters are in a fully connected neural network with 2 inputs, 3 hidden units, and 2 outputs?

(A) 13
(B) 15
(C) 17
(D) 19

Part (b) [1 MARK]

What is the primary purpose of the attention mechanism in a transformer network?

(A) To enable parallel processing of sequences
(B) To mitigate the vanishing gradient problem
(C) To reduce the number of parameters in the network
(D) To selectively focus on specific parts of the input

Part (c) [1 MARK]

In the context of RNNs, what is the primary difference between a gated recurrent unit (GRU) and an LSTM cell?

(A) LSTM uses a cell state, while GRU does not
(B) GRU has a higher computational complexity
(C) LSTM is used for sequence-to-sequence tasks, while GRU is used for sequence classification
(D) GRU has a higher risk of overfitting

Part (d) [1 MARK]

Which of the following best describes the primary goal of a generative adversarial network (GAN)?

(A) To generate new data samples that mimic the distribution of the training data
(B) To classify data samples into different categories
(C) To cluster data samples based on their similarity
(D) To learn a compressed representation of the input data

Part (e) [1 MARK]

In the context of transformers, the multi-head self-attention mechanism helps to:

(A) Reduce the overall number of parameters
(B) Encode the position information of input tokens
(C) Attend to different aspects of the input simultaneously
(D) Reduce the computational complexity

Part (f) [1 MARK]

In an autoencoder network, what is the primary purpose of the bottleneck layer?

(A) To improve training speed
(B) To obtain a compressed representation of the input data
(C) To reduce the risk of overfitting
(D) To facilitate unsupervised learning

Part (g) [1 MARK]

Which of the following functions is commonly used as a loss function for training classification models in deep learning?

(A) Mean Squared Error (MSE)
(B) Root Mean Squared Error (RMSE)
(C) Cross-Entropy Loss
(D) Cosine Similarity

Part (h) [1 MARK]

Which of the following types of information can be included as features in Graph Neural Networks to enhance their learning capabilities?

(A) Node features only
(B) Edge features only
(C) Both node and edge features
(D) Neither node nor edge features

Part (i) [1 MARK]

In the context of Convolutional Neural Networks (CNNs), what does the term "stride" refer to?

(A) The number of filters in a convolutional layer
(B) The amount by which the filter moves during each convolution operation
(C) The size of the filter used in the convolutional layer
(D) The dimensionality reduction factor applied after a pooling layer

Part (j) [1 MARK]

For a given input tensor of shape (batch_size, height, width, channels) and a filter of shape (filter_height, filter_width, channels, num_filters), which of the following correctly represents the output tensor shape after applying a "VALID" convolution in a CNN, assuming a stride of 1 and no padding?

(A) (batch_size, height - filter_height + 1, width - filter_width + 1, num_filters)
(B) (batch_size, height - filter_height, width - filter_width, num_filters)
(C) (batch_size, height + filter_height - 1, width + filter_width - 1, num_filters)
(D) (batch_size, height, width, num_filters)

Question 2. [10 MARKS]

Provided below is a confusion matrix applied to a test dataset for a three-class model used to identify cats, dogs or birds.

		Predicted		
		cats	dogs	birds
Actual	cats	30	17	20
	dogs	56	0	12
	birds	21	22	25

Part (a) [4 MARKS]

How are the samples distributed? Remark on each category. Is this a balanced dataset?

Part (b) [2 MARKS]

For which category did the model achieve the best prediction result? What percent were correctly identified?

Part (c) [2 MARKS]

Calculate the total accuracy of the model.

Part (d) [2 MARKS]

Explain why a confusion matrix can be more informative than just reporting overall accuracy for a classification problem.

Question 3. [10 MARKS]

Suppose we want to update the weights of a single neuron. The neuron models the relationship between a binary dependent variable such as a yes/no outcome. Suppose we decide that we want to update the weights of the model using the update rule

$$w_i^{t+1} = w_i^t - \gamma \frac{\partial \mathcal{L}}{\partial w_i}, \tag{1}$$

where w_i is the ith weight in the model, $\mathcal{L} = (y-t)^2 + \lambda \sum_{i=1}^{d} w_i^2$ is the loss function, $y \in \mathbb{R}$ is the model output, $t \in \mathbb{R}$ is the ground truth, and γ is the learning rate.
We can define our model as the mapping $y : \mathcal{X} \to \mathbb{R}$ where the input $\mathbf{x} = [x_1, \ldots, x_N]^T \in \mathcal{X} \in \mathbb{R}^d$. More specifically, $y(\mathbf{x}) = a(f(\mathbf{x}))$, where $f = \sum_{i=1}^{d} w_i x_i + b$, the bias term $b \in \mathbb{R}$, and the activation function $a(x) = \frac{1}{1+e^{-x}}$.

Part (a) [7 MARKS]

Find the closed form expression for the gradient $\frac{\partial \mathcal{L}}{\partial w_i}$

Part (b) [3 MARKS]

Find the closed form expression for the gradient $\frac{\partial \mathcal{L}}{\partial b}$.

Question 4. [10 MARKS]

What is the expected output of the commands below? If you expect that an error would occur, write "ERROR" without further explanation.

```
import torch.nn as nn
import torch
x1 = torch.randn(30, 16)
x2 = torch.randn(30, 16, 32, 32)
```

The shapes x1.shape and x2.shape are torch.Size([30, 16]) and torch.Size([30, 16, 32, 32]), respectively.

Part (a) [2 MARKS]

Fully connected linear layer:

```
lin = nn.Linear(in_features=16, out_features=8)
print(lin(x1).shape)
print(lin(x2).shape)
```

Part (b) [2 MARKS]

Convolutional layer:

```
conv1 = nn.Conv2d(in_channels=16, out_channels=40, kernel_size=8, padding=2)
print(conv1(x2).shape)
```

Part (c) [2 MARKS]

Convolutional layer:

```
conv2 = nn.Conv2d(in_channels=16, out_channels=32, kernel_size=16,padding=0,
    stride=4)
print(conv2(x2).shape)
```

Part (d) [2 MARKS]

Pooling layer:

```
pool = nn.MaxPool2d(kernel_size=16, stride=3, padding=4)
print(pool(x2).shape)
```

Part (e) [2 MARKS]

Convolutional transpose layer

```
convt = nn.ConvTranspose2d(in_channels=16, out_channels=8, kernel_size=3, stride
    =2, padding=1, output_padding=1)
print(convt(x2).shape)
```

Question 5. [10 MARKS]

Suppose you are tasked with building a deep learning model using a Gated Recurrent Unit (GRU) for the purpose of text generation. The model will be trained on a dataset of text to generate new sentences that follow the style and structure of the input text. Answer the following questions related to the GRU:

Part (a) [2 MARKS]

Fill in the blanks in the following Python code snippet to build a simple RNN using the PyTorch library. You may assume that all necessary packages have been imported and that the code has no syntax errors.

```
class TextGenerationGRU(nn.Module):
    def __init__(self, vocab_size, hidden_size, output_size):
        super(TextGenerationGRU, self).__init__()

        self.hidden_size = hidden_size
        self.gru = nn.GRU(input_size=_______, hidden_size=hidden_size,
num_layers=1, batch_first=True)
        self.fc = nn.Linear(hidden_size, output_size)
        self.softmax = nn.LogSoftmax(dim=1)

    def forward(self, x, hidden):
        output, hidden = self._______(x, hidden)
        output = self.fc(output[:, -1, :])
        output = self.softmax(output)
        return output, hidden
```

Part (b) [8 MARKS]

Describe the training process for a GRU-based text generation model, including how the data is prepared, how the model is trained, and the role of the loss function. Additionally, explain how the trained model can be used for generating new text.

Question 6. [10 MARKS]

In the context of Transformer networks, the concept of neural dictionaries plays a crucial role in understanding the attention mechanism. Answer the following questions related to neural dictionaries and attention in transformers:

Part (a) [6 MARKS]

Explain the role of Query (Q), Key (K), and Value (V) matrices in the attention mechanism of a Transformer network.

Part (b) [2 MARKS]

Consider the computation of the scaled dot-product attention

$$\text{attention}(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

where $Q \in \mathbb{R}^{n\times k}$, $K \in \mathbb{R}^{n\times k}$, and $V \in \mathbb{R}^{n\times v}$ matrices. What is the purpose of scaling the dot-product using $\sqrt{d_k}$, and what is the final output shape of this operation?

Part (c) [2 MARKS]

In the Transformer architecture, the self-attention mechanism is responsible for computing the relationships between different input elements. Explain how this mechanism contributes to the neural dictionary aspect of the Transformer.

Question 7. [10 MARKS]

Consider a binary classification task where the goal is to predict whether a given chemical compound is toxic or not using Graph Neural Networks (GNNs). Chemical compounds can be represented as graphs, where atoms are nodes and bonds are edges. Due to the irregular structure of chemical compounds, GNNs are particularly well-suited for this task.

Part (a) [6 MARKS]

Describe the general architecture of a GNN model for this task, mentioning the key components and their roles in the process. Include an explanation of how the GNN handles the input chemical compound graph data and how it generates the final prediction for the binary classification.

Part (b) [4 MARKS]

In the context of the chemical compound classification task, it is important to consider not only local features around each atom but also global features that capture the overall structure of the compound. Propose a modification to the GNN architecture described in Part (a) that enables the model to better capture global information about the input compound. Discuss the benefits and potential challenges of incorporating this modification into the model.

Question 8. [10 MARKS]

Consider a deep learning model that utilizes a Generative Adversarial Network (GAN) for generating realistic synthetic data. Answer the following questions related to the GAN model:

Part (a) [4 MARKS]

Describe the architecture and main components of a GAN, including the roles of the generator and discriminator networks. Explain how these two networks interact during the training process.

Part (b) [3 MARKS]

Explain the concept of mode collapse in the context of GANs. Discuss its implications on the quality of generated samples and suggest one method to mitigate the problem of mode collapse.

Part (c) [3 MARKS]

How do we define the loss function for the generator? How do we define the loss function for the discriminator?

Question 9. [10 MARKS]

Deep learning models, such as GNNs, Transformers, GANs, RNNs, LSTMs, GRUs, CNNs, and ANNs, share several common features and challenges during training. This question explores these commonalities in the context of training deep neural networks. Please answer the following questions below:

Part (a) [3 MARKS]

Identify and briefly explain three common features or architectural elements present in the majority of the deep learning models mentioned above. Discuss their roles and how they contribute to the learning capabilities of these models.

Part (b) [3 MARKS]

During the training process, deep learning models often face various challenges or issues. Identify and briefly explain three common challenges encountered when training deep neural networks, regardless of the specific model architecture.

Part (c) [4 MARKS]

Compare and contrast the training process of a Convolutional Neural Network (CNN) and a Transformer network. Discuss the main differences in terms of architecture, model type, and dataset requirements, as well as any similarities in the steps involved in the training process. Provide at least two differences and one similarity between these two types of networks.

Total Marks = 90

END OF FINAL EXAM